COMPUTER NETWORKING NOTES

Cybersecurity Fundamentals — Day 1 & Day 2

DAY 1: INTRODUCTION TO NETWORKING

1. What Is Networking?

Networking is the practice of connecting two or more computers or devices to communicate and share data, hardware, services, and other resources.

Example

A Wi-Fi router connects a computer, mobile phone, smart TV, and printer to a network.

Why Is Networking Important in Cybersecurity?

Cybersecurity professionals must understand how data moves between devices and networks to identify vulnerabilities and protect systems from attacks.

2. Types of Networks

1. PAN — Personal Area Network

A network that connects personal devices over a short distance.

Example: A mobile phone connected to Bluetooth headphones.

2. LAN — Local Area Network

A network that connects devices within a limited geographical area.

Example: A home, office, or school network.

3. WLAN — Wireless Local Area Network

A LAN that connects devices wirelessly, usually through Wi-Fi.

Example: A Wi-Fi network at home.

4. CAN — Campus Area Network

A network that connects multiple LANs within a campus or organization.

Example: A university campus network.

5. MAN — Metropolitan Area Network

A network that connects multiple networks across a city or metropolitan area.

Example: A network connecting offices in different parts of a city.

6. SAN — Storage Area Network

A specialized network that provides servers with access to shared storage devices.

Example: Enterprise data-center storage.

7. WAN — Wide Area Network

A network that connects devices or smaller networks across large geographical areas.

Example: A company's network connecting offices in different countries. The Internet is the best-known example of a global network.

Quick Revision

- PAN — Personal Area Network
- LAN — Local Area Network
- WLAN — Wireless Local Area Network
- CAN — Campus Area Network
- MAN — Metropolitan Area Network
- SAN — Storage Area Network
- WAN — Wide Area Network

DAY 2: NETWORK DEVICES

Network devices are physical hardware components that help computers and other devices communicate with one another.

1. Router — OSI Layer 3

A router connects different IP networks and forwards data packets using IP addresses and a routing table.

Main functions:

- Connects a local network to other networks.
- Forwards packets using IP addresses.
- Selects routes for data transmission.
- Commonly connects a home network to the Internet.

Example: A home Wi-Fi router.

2. Switch — OSI Layer 2

A switch connects multiple devices within a local area network (LAN).

Main functions:

- Forwards Ethernet frames using MAC addresses.
- Learns which MAC addresses are reachable through its ports.
- Maintains a MAC address table.
- Reduces unnecessary traffic compared with a hub.

Example: Connecting computers in an office.

3. Hub — OSI Layer 1

A hub connects multiple Ethernet devices but sends incoming signals to all other connected ports.

Main functions:

- Operates at the Physical layer.
- Broadcasts incoming data to the other ports.
- Does not intelligently identify the destination device.
- Can create unnecessary network traffic.

Note: Hubs are largely obsolete and have mostly been replaced by switches.

4. Modem

A modem connects a customer's network to an Internet Service Provider (ISP) using the provider's access technology.

Main functions:

- Converts or modulates signals as required by the connection technology.
- Enables communication over the ISP's access network.
- Provides the connection used for Internet access.

Note: A modem and a router perform different functions, although many modern devices combine both.

5. Repeater — OSI Layer 1

A repeater regenerates or repeats signals to extend the distance over which data can travel.

Main functions:

- Extends network communication distance.
- Regenerates weakened signals.

- Does not route packets using IP addresses.

Example: A Wi-Fi range extender.

6. Bridge — OSI Layer 2

A network bridge connects or separates LAN segments and forwards frames based on MAC addresses.

Main functions:

- Connects network segments.
- Filters traffic using MAC addresses.
- Reduces unnecessary traffic between segments.

Note: A network switch is essentially a multiport bridge.

7. Gateway

A gateway connects networks or systems and may translate between different protocols or communication formats.

Main functions:

- Acts as an entry or exit point between networks.
- Can translate between protocols.
- Allows communication between different systems.

Example: In a typical home network, the default gateway is usually the router used to reach external networks.

8. Firewall

A firewall is a hardware device, software program, or service that monitors and controls incoming and outgoing network traffic according to security rules.

Main functions:

- Allows or blocks network traffic.
- Enforces security policies.
- Can inspect IP addresses, ports, protocols, and application traffic.
- Helps protect networks from unauthorized access.

A firewall is an important component of network security.

9. NIC — Network Interface Card or Controller

A Network Interface Card (NIC) allows a computer or device to connect to a network.

Main functions:

- Provides a wired Ethernet or wireless network interface.
- Enables sending and receiving network data.
- Usually has a MAC address.

Example: A computer's Ethernet adapter or Wi-Fi adapter.

10. Proxy Server

A proxy server acts as an intermediary between a client and another server.

Main functions:

- Forwards requests on behalf of clients.
- Can filter requests and restrict access.
- May cache content to improve performance.
- Can log network activity according to its configuration.

Example: A company proxy used to manage employees' web access.

11. Load Balancer

A load balancer distributes requests or network traffic across multiple servers.

Main functions:

- Distributes workloads.
- Improves availability and reliability.
- Helps prevent one server from becoming overloaded.
- Can improve application performance.

Example: Distributing website visitors across several web servers.

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3. NETWORK TOPOLOGIES

A network topology describes how network devices (nodes) and connections (links) are arranged.

1. Bus Topology

In a bus topology, all devices connect to a single main backbone cable.

Advantages:

- Simple design.
- Can be inexpensive for small networks.

Disadvantages:

- A backbone failure can disrupt the network.
- Performance may decline as traffic increases.

2. Star Topology

In a star topology, every device connects to a central device, usually a switch.

Advantages:

- Easy to install and manage.
- A single device-link failure usually affects only that device.

Disadvantages:

- Failure of the central switch can disconnect the attached devices.
- Requires separate links to the central device.

3. Ring Topology

In a ring topology, each device connects to two neighboring devices, forming a closed loop.

Advantages:

- Provides an organized path for data transmission.
- Can offer predictable traffic behavior in some implementations.

Disadvantages:

- A link or device failure can disrupt a basic ring unless protection mechanisms are used.

4. Mesh Topology

In a mesh topology, devices have multiple interconnections, providing alternative paths for communication.

Advantages:

- High redundancy.
- Alternative paths may remain available when a link fails.
- Can improve reliability.

Disadvantages:

- More expensive to implement.
- Requires additional connections and configuration.

5. Tree Topology

A tree topology is a hierarchical network structure with parent devices and branching connections.

Advantages:

- Suitable for large, organized networks.
- Easier to expand by adding branches.

Disadvantages:

- Failure of an important upstream device can affect an entire branch.

6. Hybrid Topology

A hybrid topology combines two or more different topology types, such as star, bus, ring, or mesh.

Advantages:

- Flexible design.
- Can be adapted to organizational needs.

Disadvantages:

- May be more complex to design and maintain.

Quick Revision

- Bus — One main backbone cable.
- Star — Central switch or hub.
- Ring — Closed loop.
- Mesh — Multiple interconnected paths.
- Tree — Hierarchical branches.
- Hybrid — Combination of topologies.

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4. DATA AND NETWORK COMMUNICATION

What Is Data?

Data refers to facts or information represented in a form that computers can store, process, or transmit.

Examples: Text, images, audio, video, and files.

IP Address

An IP address is a logical address used to identify a network interface for communication across IP networks.

Example: 192.168.1.10

MAC Address

A MAC address is a link-layer address associated with a network interface and commonly used to deliver Ethernet frames within a local network.

Example: 00:1A:2B:3C:4D:5E

Frame

A frame is a data unit used at the Data Link layer. An Ethernet frame contains source and destination MAC addresses.

Packet

A packet is a data unit at the Network layer. An IP packet contains source and destination IP addresses.

Physical Layer

The Physical layer transmits bits through electrical, optical, or radio signals using the relevant physical medium.

Examples: Ethernet cables, fiber-optic cables, and Wi-Fi radio signals.

Simple Layer Relationship

- Network layer — IP packets and IP addresses.
- Data Link layer — Frames and MAC addresses.
- Physical layer — Bits, signals, and transmission media.

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5. SWITCH MAC ADDRESS TABLE

A switch learns source MAC addresses from incoming Ethernet frames and associates them with the ports on which those addresses were seen.

Example:

MAC Address	Switch Port
AA:AA:AA:AA:AA:01	Port 1
BB:BB:BB:BB:BB:02	Port 2
CC:CC:CC:CC:CC:03	Port 3

When the switch receives a frame, it checks the destination MAC address in its MAC address table. If the destination is known, it forwards the frame through the appropriate port. If the destination is unknown, it normally floods the frame to other ports in the same VLAN, excluding the incoming port.

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6. OSI MODEL — SEVEN LAYERS

The OSI model is a conceptual framework used to understand how network communication works.

Layer	Name	Main Function
7	Application	Network services used by applications
6	Presentation	Data representation, encoding, encryption
5	Session	Establishes and manages communication sessions
4	Transport	End-to-end delivery, segmentation, reliability
3	Network	Logical addressing and routing
2	Data Link	Frames, MAC addresses, local delivery
1	Physical	Bits, signals, cables, and radio transmission

Important Network Devices by Layer

- Layer 1 — Hub, repeater.
- Layer 2 — Switch, bridge.
- Layer 3 — Router.

These are common classifications. Some modern devices combine functions across multiple layers.

Memory Trick

Top to Bottom: All People Seem To Need Data Processing.

Application → Presentation → Session → Transport → Network → Data Link → Physical.

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7. IMPORTANT CYBERSECURITY REVISION POINTS

1. Networking enables devices to communicate and share resources.
2. A router forwards IP packets between networks.
3. A switch forwards Ethernet frames using MAC addresses.
4. A hub sends incoming signals to its other ports.
5. A firewall enforces network security rules.
6. A NIC connects a device to a network.
7. A proxy server acts as an intermediary.
8. A load balancer distributes traffic across servers.
9. IP addresses are logical addresses; MAC addresses are link-layer addresses.
10. Understanding networking is essential for cybersecurity, vulnerability assessment, and penetration testing.